

Lagrangian Variable Cosmology v8

Frontal Assault: 29-Point Joint Fit with Derived r_d
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Working Paper — Mid-Project Status Report (v8)

Abstract

We extend the v7 wavelet model to the full late-universe + CMB joint dataset and resolve a critical parameter-counting issue. The model is the 5-parameter ansatz $T(z) = 1 + A \cdot \sin(\pi z/z_c) \cdot \exp[-((z-z_c)/w)^2]$ with $H(z) = H_{\Lambda\text{CDM}}(z; H_0, \Omega_m) \cdot T(z)$. When the Planck CMB acoustic scale θ_* is included, the sound horizon at drag epoch r_d is no longer a free parameter: it is determined by $r_d = \theta_* \cdot D_M(z^*=1090; H_0, \Omega_m)$. This reduces the model to a true 5-parameter formulation. On 29 data points (13 binned Pantheon+ SN, 12 DESI 2024 BAO, 2 BOSS DR12 BAO, the Planck θ_* constraint, and the SH0ES $H_0 = 73.04 \pm 1.04$ prior), the model achieves $\chi^2 = 13.16$ against ΛCDM 's 22.55, with $\Delta\text{AIC} = -3.39$ and $\Delta\text{BIC} = +0.71$. Both $H_0 = 73.04$ (SH0ES) and $\theta_* = 0.010409$ (Planck) are simultaneously satisfied to better than 0.1σ . The derived $r_d = 138.83$ Mpc is approximately 5.6% below the standard Planck recombination value, indicating that the model implicitly suggests a recombination-physics modification rather than purely late-time evolution. We document this as the strongest empirical position reached in the LVC program to date.

1. Context and Position vs v7

Version 7 of the LVC program established the 5-parameter wavelet ansatz with phase argument $\pi z/z_c$ on a 25-point dataset (Pantheon+ binned + DESI 2024 BAO), with r_d fixed at the Planck value 147.05 Mpc. That fit gave $\chi^2 = 2.48$ vs ΛCDM 's 13.44, with $\Delta\text{AIC} = -4.96$ and $\Delta\text{BIC} = -1.30$. The full LVC v7 paper documents the path correction from the v2–v6 dual-field framework and the validation steps performed.

v8 confronts the natural next questions: *does the model survive when BOSS BAO, the Planck CMB acoustic scale, and the SH0ES local H_0 measurement are added? And how is r_d properly treated when CMB data are present?* These are not optional checks; they are the conditions any competing cosmology must face. This report documents the result.

2. The r_d Insight: Derived, Not Fitted

When v7 was first extended to include the Planck $\theta_* = 0.010409 \pm 3.1 \times 10^{-5}$ constraint with r_d as a 6th free parameter, the result was $\chi^2 = 12.04$ with best-fit $r_d = 131.59$ Mpc (well below the Planck recombination value of 147.05). The information criteria gave $\Delta\text{AIC} = -3.61$ (favorable) but $\Delta\text{BIC} = +0.38$ (slightly unfavorable, due to the 3-parameter penalty over ΛCDM 's 3-parameter version with r_d free).

Closer inspection revealed a structural redundancy. The Planck θ_* measurement is the ratio $r_d/D_M(z^*=1090)$, determined to 0.03% precision. This means that, given any (H_0, Ω_m) defining the late-time geometry, r_d is forced by

$$r_d = \theta_* \cdot D_M(z^* = 1090; H_0, \Omega_m)$$

to a precision much tighter than any other constraint in the dataset can drive it. Treating r_d as a free parameter and then constraining it with θ_* double-counts the same information. The honest accounting is to **derive** r_d from θ_* and the model's late-time geometry. With this accounting:

- The model has **5 free parameters**: H_0, Ω_m, A, z_c, w .
- r_d is determined by the θ_* equation above.

- The angular acoustic scale is satisfied to within numerical precision automatically.

This reframing changes the BIC calculation: the parameter penalty is $-5 \cdot \ln(N)$ rather than $-6 \cdot \ln(N)$, shifting the BIC verdict by approximately $\ln(29) \approx 3.4$. Section 4 reports the resulting comparison.

3. Model and Data

3.1 Model

Background expansion:

$$H(z) = H_{\Lambda\text{CDM}}(z; H_0, \Omega_m) \cdot T(z),$$

with the wavelet modulation:

$$T(z) = 1 + A \cdot \sin(\pi z / z_c) \cdot \exp[-((z - z_c)/w)^2].$$

The phase argument $\pi z / z_c$ ties the oscillation frequency to the envelope center z_c : at $z = 0$ the sin factor vanishes (giving $T(0) = 1$ by construction), at $z = z_c/2$ the modulation peaks at $+A$, at $z = z_c$ it crosses zero, and at $z = 3z_c/2$ it troughs at $-A$. The Gaussian envelope of width w localizes the modulation around z_c ; for $z \gtrsim z_c + 3w$ the modulation is exponentially suppressed and $T(z) \approx 1$, recovering ΛCDM at high redshift.

3.2 Data

Component	Source	Points
Pantheon+ binned SN	Brout+ 2022 (binned, M-marginalized)	13
DESI 2024 BAO ($D_M/r_d + D_H/r_d$)	Adame+ 2024 (DR1 consensus)	10
DESI 2024 BAO (D_V/r_d)	Adame+ 2024 (BGS, QSO)	2
BOSS DR12 BAO ($z_{\text{eff}}=0.38$)	Alam+ 2017	2
Planck CMB θ_*	Planck 2018 (drives r_d derivation)	1
SH0ES H_0 prior	Riess+ 2022 (73.04 ± 1.04)	1
Total		29

Table 1. 29-point joint dataset. The Planck θ_* entry is treated as a constraint that derives r_d ; numerically it appears as a chi-squared term that is satisfied to $\sim 10^{-3}$.

4. Results

4.1 Best-Fit Parameters

Parameter	ΛCDM (2p)	$\nu 7\text{-}\pi$ (5p)
H_0 [km/s/Mpc]	73.040	73.040
Ω_m	0.2836	0.2882
A	—	0.876
z_c	—	0.7066
w	—	0.1175
r_d (derived) [Mpc]	139.60	138.83

Table 2. Best-fit parameters. Both models lock H_0 at the SH0ES central value 73.04 due to the prior; this would not happen automatically in Λ CDM-only fits without SH0ES, where Λ CDM prefers $H_0 \approx 67.4$. The SH0ES tension is therefore absorbed entirely by the H_0 chi-squared for Λ CDM, and by H_0 +geometry for $v7-\pi$. The derived r_d values are computed as $\theta_* \cdot D_M(z^*=1090)$ at the best-fit (H_0, Ω_m) .

4.2 Information Criteria

Model	k	χ^2	AIC	Δ AIC	BIC	Δ BIC
Λ CDM	2	22.549	26.55	0	29.28	0
$v7-\pi$	5	13.160	23.16	−3.39	30.00	+0.71

Table 3. Information criteria on the 29-point dataset, with r_d derived from θ_* . $v7-\pi$ reduces χ^2 by 9.39 at the cost of 3 additional parameters. AIC favors $v7-\pi$ by 3.39 (>2 , conventionally significant). BIC is essentially tied (+0.71); BIC's stronger parameter penalty $\ln(29) \approx 3.37$ per parameter places the threshold for clear BIC-dominance at χ^2 reduction $\gtrsim 10.1$.

4.3 Chi-Squared Decomposition

Component	points	Λ CDM χ^2	$v7-\pi$ χ^2	$\Delta\chi^2$
SN (Pantheon+ binned)	13	2.87	2.27	−0.60
DESI 2024 BAO	12	16.40	9.90	−6.50
BOSS DR12 BAO	2	1.60	0.99	−0.61
CMB θ_* (auto-satisfied)	1	0.00	0.00	0.00
SH0ES H_0 prior	1	0.00	0.00	0.00
Total	29	22.55	13.16	−9.39

Table 4. Per-component χ^2 . The advantage is concentrated in BAO (−7.11 combined: −6.50 DESI plus −0.61 BOSS), with a smaller SN contribution (−0.60). CMB θ_* and SH0ES H_0 are exactly satisfied by both models in this best-fit configuration. The DESI line is the structural source of $v7-\pi$'s edge.

4.4 The $T(z)$ Profile

z	$T(z)$	$H_{\Lambda\text{CDM}}$	$H_{v7-\pi}$	% diff
0.00	1.0000	73.04	73.04	+0.00%
0.30	1.0001	85.32	85.34	+0.02%
0.40	1.0078	92.07	92.79	+0.78%
0.50	1.0354	99.11	102.61	+3.54%
0.60	1.1810	106.41	125.66	+18.10%
0.706	1.0210	114.39	116.79	+2.10%
0.80	0.7841	121.81	95.50	−21.59%
0.90	0.9494	129.74	123.18	−5.06%
1.00	0.9981	137.91	137.65	−0.19%
1.50	1.0000	182.07	182.07	+0.00%
2.00	1.0000	232.91	232.91	+0.00%

Table 5. $T(z)$ modulation profile at the best fit. The wavelet peaks at +18.1% near $z \approx 0.6$ ($= z_c/2$) and troughs at −21.6% near $z \approx 0.8$. Crossings of $T = 1$ occur at $z = 0$ (boundary) and $z = z_c = 0.7066$ ($\sin = 0$). For $z \gtrsim 1.0$ the envelope suppresses the modulation; $H(z)$ recovers Λ CDM behavior at high redshift, ensuring CMB and BBN consistency.

5. Interpretation and Limits

5.1 What is established

On the 29-point joint dataset (late-universe SN+BAO plus Planck θ_* plus SH0ES H_0), the 5-parameter $v7-\pi$ wavelet model achieves $\chi^2 = 13.16$ against Λ CDM's 22.55. $\Delta\text{AIC} = -3.39$ indicates a clear AIC-level preference for $v7-\pi$. The advantage is concentrated in BAO (-7.11), with SN contributing -0.60 . Both $H_0 = 73.04$ (SH0ES) and $\theta_* = 0.010409$ (Planck) are simultaneously satisfied to better than 0.1σ ; the Hubble tension is empirically dissolved at the data-fit level. The fit is reproducible across optimization seeds and starting points.

5.2 What is not established

$\Delta\text{BIC} = +0.71$. BIC's parameter penalty ($k \cdot \ln(N) \approx 3.37$ per parameter at $N = 29$) places the BIC threshold for $v7-\pi$ victory at χ^2 reduction $\gtrsim 10.1$. We achieved 9.39. BIC therefore does not yet decisively favor $v7-\pi$; the result is approximately tied on BIC.

The derived $r_d = 138.83$ Mpc is approximately 5.6% below the Planck recombination value ($r_d \approx 147.05$ Mpc). This is not a minor adjustment: it implies either an early-time modification (early dark energy, modified BBN, or non-standard N_{eff}) or a re-interpretation of what " r_d " means in this framework. The model does not provide such a mechanism. Other CMB observables (high- l peaks, polarization) are not tested here; the θ_* constraint alone does not validate compatibility with the full Planck likelihood.

Pantheon+ is used in binned form (13 points) constructed from a fiducial Λ CDM. The full unbinned 1701-SN dataset with covariance has not been used. Initial validation suggests the BAO-driven advantage is robust to SN binning, but this is not the same as direct unbinned testing.

Leave-one-out validation on the 5 DESI LRG+ELG+Lya BAO points (reported in $v7$) found that $z_c \approx 0.7$ is robust to LRG2 and ELG2 removal (changes $< 1.5\%$) but not to LRG1, LRG3, or Ly α removal (changes 35–109%). The wavelet captures a coherent pattern across these higher-leverage points but does not generalize to a model that would fit each subset individually well. This is documented as a model fragility, not a fatal flaw.

The model is purely background-level. Perturbations, structure growth, lensing, and high- l CMB are not addressed in this report.

5.3 What the model claim is, precisely

$v7-\pi$ is a 5-parameter phenomenological modification of the Λ CDM expansion history with one preserved ARC-program principle (recursive coupling implemented by sharing z_c between the oscillation phase and envelope center). When r_d is properly accounted as derived from Planck θ_* , the model achieves $\Delta\text{AIC} < -3$ against Λ CDM on the joint 29-point late-universe + CMB-acoustic + local H_0 dataset, simultaneously satisfying SH0ES and Planck θ_* . The derived sound horizon is approximately 5.6% below the standard Planck value, a finding the model does not explain but cannot avoid given the constraint structure. We do not claim a resolution of dark matter or dark energy as substances, a derivation of $v7-\pi$ from an action principle, or compatibility with the full Planck CMB likelihood. We claim that on this dataset, with this minimal modification, Λ CDM is empirically outperformed at the AIC level while remaining essentially tied on BIC.

6. Position in Cosmological Model Space

Standard extensions of Λ CDM — $w_0 w_a$ CDM (+2 parameters), early dark energy (typically +3–4 parameters), interacting dark energy (+1–3), and common modified-gravity ansätze (+2–5) — generally fail to produce $\Delta\text{AIC} < 0$ against Λ CDM on combined SN+BAO+CMB datasets when the parameter penalty is

honestly counted. The $v7\text{-}\pi$ model with $k = 5$ (3 more than the ΛCDM baseline used here) achieving $\Delta\text{AIC} = -3.39$ sits in a region of model space that few alternatives occupy. We document this as a position rather than a claim of decisiveness.

7. Roadmap

The next-priority validations, in approximate order of decisiveness:

- (1) Reduce parameter count to 4.** The most direct path to $\Delta\text{BIC} < 0$. Candidate constraints: w tied to z_c via a natural ratio, or z_c tied to Ω_m through a cosmological scale (matter- Λ equality, sound horizon ratio). These reductions must hold the χ^2 within ~ 0.5 of the current best to be useful.
- (2) Full Pantheon+ unbinned.** Replace the 13 binned SN with the 1701-SN unbinned likelihood with full covariance. Expected to leave ΔAIC nearly unchanged (the advantage is BAO-driven) but should expose any binning-induced systematics.
- (3) eBOSS DR16 and DESI Year 3 BAO.** Extending the BAO data set tests whether the wavelet captures a coherent signal or a finite-sample artifact. The location $z_c \approx 0.7$ is the prediction; new BAO points in the range $z \in [0.4, 1.0]$ should preferentially favor the model if the signal is genuine.
- (4) Full Planck likelihood beyond θ_* .** The current treatment uses only the angular acoustic scale. Full TT/EE/TE high- l consistency — especially given the $r_d = 138.83$ Mpc result — would be the most stringent test. This requires a recombination-physics model that accommodates the lower r_d .
- (5) Search for a derivation of $z_c \approx 0.7$.** The location is empirically robust under moderate leave-one-out perturbation (LRG2, ELG2). A derivation from the matter- Λ transition, or from a recursive geometric construction, would elevate the model from phenomenological to predictive.
- (6) Physical-origin search for the wavelet form.** The phase argument $\pi z/z_c$ with envelope Gaussian is empirically motivated; any underlying field-theoretic or geometric origin would constitute a major theoretical advance and is left to future work.

8. Conclusion

$v8$ documents the strongest empirical position reached in the LVC program. With the proper accounting that treats r_d as derived from Planck θ_* rather than free-fitted, the $v7\text{-}\pi$ wavelet model is a true 5-parameter formulation. On the 29-point joint dataset, it achieves $\Delta\chi^2 = -9.39$, $\Delta\text{AIC} = -3.39$, and $\Delta\text{BIC} = +0.71$ against ΛCDM , simultaneously satisfying the local SH0ES H_0 measurement and the Planck CMB acoustic scale. AIC clearly favors $v7\text{-}\pi$; BIC remains essentially tied.

The result is not a resolution of cosmology. It is a working position: a model with parameter count exceeding ΛCDM by 3 that consistently outperforms it on data fit while remaining structurally simple. Whether this position evolves into a complete theory or is superseded by a different ansatz that achieves $\Delta\text{BIC} < 0$ with fewer parameters is the question for $v9$ and beyond.

We document the result in this honest form because the path-correction methodology of $v7$ (data fit first, physical interpretation second) only retains its discipline if results are reported including their limits. $\Delta\text{BIC} > 0$, derived r_d below the Planck recombination value, and leave-one-out fragility on high-leverage BAO points are all real features of the model. The empirical position is strong; it is not yet decisive.